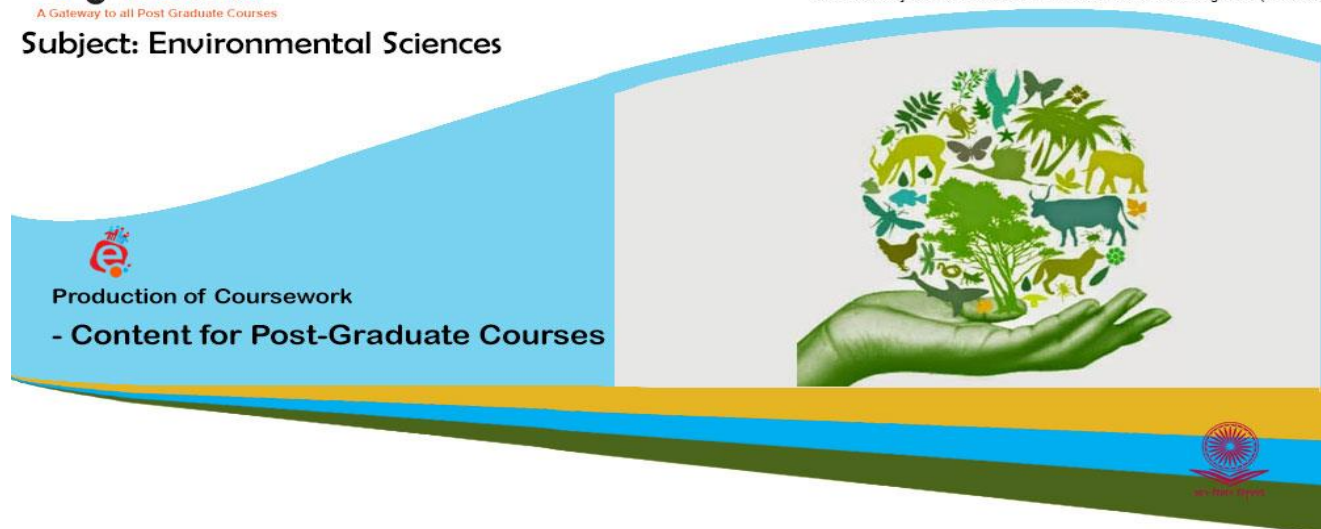


Subject: Environmental Sciences



Production of Coursework

- Content for Post-Graduate Courses

Paper No: 13 Environmental Law and Policies

Module: 32 Climate Change



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Description of Module	
Subject Name	Environmental Sciences
Paper Name	Environmental Law and Policies
Module Name/Title	Climate Change
Module Id	EVS/ELP-XIII/32
Pre-requisites	
Objectives	Reader will be apprised in brief about the concept of Climate Change, the effect of climate change and its impact on human beings. Learners will learn about the international efforts made to combat, mitigate and adapt to climate change and India's efforts in this direction
Keywords	Climate Change, NASA, USGCRP

Climate Change

Introduction

Climate change is an important and serious issue confronting the mankind which adversely affects the survival of human beings on this planet. The issue has become serious due to unregulated human activities which have resulted in change in the usual weather found at a particular place. Whereas changes in weather can be felt by every person because they are visible as the weather changes frequently over one day or over few hours, changes in climate can not be seen/felt as such in short span, because it changes over years i.e. over large span of time.

Climate change is referred to as the changes in the ordinary weather found at a particular place over a period of time e.g. the usual weather, rainfall in rainy season, changes in winter season and spells, changes in summer season and spells etc in a particular place. Climate change also refers to changes in earth's climate in totality i.e. changes in earth's usual temperature (NASA).

As per Intergovernmental Panel on Climate Change (IPCC), Climate Change refers to change in the state of climate that can be identified (e.g. using statistical tests) by changes in the mean and/or the variability of its properties, and that persists for an extended period, typically decades or longer (IPCC Report, Fourth Assessment Cycle)

In United National Framework Convention on Climate Change (UNFCCC), Climate change is referred to as change of climate which is attributable to human activities which may directly or indirectly alter the composition of global atmosphere. As per UNFCCC, this change in climate induced by human activities should be in addition to natural variations in climate (UNFCCC, Article 1, para 2)

The change in climate is not only a rhetoric but has been established scientifically by various studies/research. As per US Global Change Research Programme Report, 2017, annual average surface air temperature has shown an increase of about 1.8°F (1.0°C) globally over the last 115 years (1901–2016). Further, the report states that the current period is the warmest in the history of modern

civilization and possibly the trend in climate change is to continue over a period of time. (USGCRP, 2017)

Learning Outcomes

In this backdrop, reader will be apprised in brief about the concept of Climate Change, the effect of climate change and its impact on human beings. Learners will learn about the international efforts made to combat, mitigate and adapt to climate change and India's efforts in this direction. The topic will be discussed under the following headings

- Climate Change : Meaning
- Effects of Climate Change
- Impact on Human Beings
- Global Attention towards Climate Change
- International Efforts to Combat, Mitigate and Adapt to Climate Change
 - Montreal Protocol
 - IPCC
 - UNFCCC
 - Kyoto Protocol
 - Paris Agreement
- Indian Efforts
 - National Action Plan on Climate Change
- Conclusion

Effects of Climate Change

Scientific studies, research and various reports presented by NASA, IPCC, USGCRP and UK Risk Assessment Reports have identified and analysed various effects of climate change including the following :

Global Temperature Rise – As stated earlier that global average surface air temperature has increased by around 1.8°F (1.0°C) globally during the last 115 years. This change has been attributed largely to emissions of carbon dioxide and other emissions (NASA). The recent period has been declared to be warmest in human history. As per NASA, since 2001, 16 out of 17 years are the warmest years on record in human history.

Shrinking Ice Sheets – Climate Change leading to global warming has led to shrinking of ice sheets. As per NASA, ice mass in Greenland and Antarctic ice sheets has decreased. NASA reports reduction of about 150 to 250 cubic Kms ice per annum between 2002 and 2006. Similarly, it reports that in Antarctica reduction of about 152 cubic kms. during 2002 to 2005 has taken place. (NASA). Arctic land and sea ice loss has also been reported by USGCRP, 2017. Average Arctic sea ice has decreased around 3.5 to 4.1% per decade and has become thinner by between 4.3 and 7.5 feet (USGCRP, 2017)

Glacial Retreat – One of the adverse impacts of climate change is the retreat of glaciers across the globe i.e. in Himalayas, Alaska, Andes, Alps etc. It is claimed that Alaska glaciers have lost mass especially after 1984 where every subsequent year has shown ice mass lesser than the preceding year.

Decreased Snow Cover – Satellite analysis has shown that snow cover has gradually decreased over the past five decades and that the snow is melting faster (NASA)

Ocean Acidification – Oceans are sinks of carbon dioxide and have been absorbing excess heat caused by GHG **emissions**. More than quarter of the CO₂ emitted in the atmosphere is absorbed by the oceans leading of warming of ocean waters at almost all levels and acidification of the oceans which is detrimental to marine ecosystems (USGCRP, 2017). Since the onset of industrial revolution, acidity of surface waters have increased by 30% (NASA) which is attributable to CO₂ emissions which are being absorbed by the oceans. The absorption of CO₂ by oceans is increasing every year making them more acidic. Absorption of CO₂ by oceans is increasing by 2 billion tons per annum.

Rise in Sea Level – One of the most worrisome impacts of climate change and global warming is the increase in sea level. Global warming, melting glaciers, shrinking ice sheets have led to increase in sea

level. It is projected that there has been an increase of 7-8 inches in global mean sea level since 1900. This rise in sea level is reported to have accelerated since 1993 and the trend of increase in sea level is likely to continue in future also. (USGCRP, 2017)

Other Effects – This apart, climate change has affected pattern of winds, pattern of precipitation and ocean circulation. This apart, oxygen concentrations at intermediate depths in oceans have shown a decline.

Impact on Human Beings

The above effects of climate change are likely to affect one and all in varying degrees. The likely impact of climate change on human beings may differ from place to place. Climate Change is likely to have the following adverse effects on humans

Flooding of Coastal Areas and Coastal Change – Rising sea levels caused by melting of glaciers and polar ice sheets and further aggravated by increase in sea temperature at all levels is likely to cause subsidence i.e. sinking of land under sea. Various coastal cities/States are facing this serious issue such as Maldives, Bangladesh, Japan, China, UK, France, India etc. Rising sea level not only leads to subsidence but is also likely to enter freshwater resources and the entry of salt water is likely to contaminate freshwater resources causing serious issue of scarcity of water for human needs. The present scale of climate change may affect huge population of the world living in coastal cities. Further, climate change may lead to increase in precipitation further increasing chances of flooding and erosion of coastline thereby causing damage to life and property.

Risks to Health, Well Being and Productivity – As stated above, climate change is likely to have huge impact on human beings. It will pose serious risk to human life, health and well being. Demographic change caused by Climate change may lead to overheating and related health issues leading to increase in heat related deaths. Overheating is likely to affect human productivity as well. (UK Climate Change Risk Assessment, 2017)

Shortages in Public Water Supply – Increasing sea level, subsidence and intrusion of salt water in freshwater lakes coupled with increasing population will put more strains on available water resources.

It is likely to cause shortage in water supply needed for various purposes thereby adversely affecting one and all. The problem will be more severe in coastal cities/states.

Effect on Food Production – Extreme weather changes caused by climate change may affect food production adversely (UK Climate Change Risk Assessment, 2017). Further, the subsidence of low lying areas like Bangladesh having fertile plains may lead to scarcity of food stuffs

Further, there will be emergence of new diseases and pests caused by climatic changes which can adversely affect humans, animals, plants and the environment.

Global Attention towards Climate Change

The following is brief timeline of events wherein global attention towards climate change was attracted leading to negotiation and adoption of UNFCCC.

- In 1859 John Tyndall brought to notice that changes in the concentration of gases could bring about climate change when he discovered that some gases block infrared radiation
- In 1896 Svante Arrhenius concluded that industrialization and coal burning will enhance the natural greenhouse effect. The first calculation of global warming from human emissions of CO₂ was published
- In 1938, Guy Stewart Callendar (British) developed a theory linking rising temperature to rising CO₂ concentrations in the atmosphere. He pointed out that post industrialization, CO₂ concentration has increased. He used data from 147 weather stations from around the world.
- In 1955, Gilbert Plass (US Researcher) developed Carbon Dioxide theory and analysed infrared absorption of various gases
- In 1957, Roger Revelle discovered that the entire CO₂ produced by humans will not be absorbed by oceans
- In 1958, US Scientist Charles David Keeling started measuring atmospheric CO₂ at Mauna Loa, Hawaii – Keeling Curve
- Wallace Broecker in 1975 coined the term Global Warming in his paper “Climate Change : Are we on the brink of a Pronounced Global Warming”

- Adoption of Montreal Protocol to Vienna Convention in 1987
- Setting up of Intergovernmental Panel on Climate Change (IPCC) in 1988

International Efforts to Contain and Combat Climate Change

The international community became alive to climate change and its likely impact on the plant and humans especially in the late twentieth century. Accordingly, various efforts were made to contain, combat and mitigate the effects of climate change. These can be broadly discussed under the following heads

- Montreal Protocol to Vienna Convention
- Setting up of IPCC
- Adoption of UNFCCC

Montreal Protocol to Vienna Convention

Vienna Convention for the protection of ozone layer was signed in 1985. The Convention was the result of the efforts of the world community to protect ozone layer when a huge hole was detected in it over Antarctica. Vienna Convention was aimed at promoting cooperation among nations regarding the depletion of ozone layer and to promote research and information exchange on the impact of human activities on ozone layer. However, the Convention was not a binding document and it provided for adoption of Protocols of binding nature for reducing the emission of Ozone Depleting Substances (ODS). To have a legally binding instrument, Montreal Protocol to Vienna Convention was adopted in 1987 which is a supplementary agreement to the Protocol. Montreal Protocol aimed at controlling and curbing global emission of substances that deplete the Ozone layer or modify the Ozone Layer and ultimately elimination of ODS. Though originally the Protocol was adopted to protect the Ozone Layer yet the potential effects of emission of ODS on climate change have been recognized since ODS are also potent greenhouse gases and contribute significantly to climate change. So far Protocol has been amended six times i.e. London Amendment, 1990; Copenhagen, 1992; Vienna Accord, 1995; Montreal Amendment, 1997; Beijing Amendment, 1999 and; Kigali Amendment, 2016. These amendments have added list of controlled substances which are required to be phased out for

protection of Ozone layer. Kigali amendment added HFC's as a controlled substance and provided schedule for reduction of use of HFCs by 80-85%. HFC phasing out is expected to prevent 105 million tonnes of carbon dioxide equivalent of greenhouse gases and will help to avoid 0.5 degree temperature rise by 2100 (UNEP).

Intergovernmental Panel on Climate Change (IPCC)

IPCC was set up in 1988 by World Meteorological Organization and United Nations Environment Programme (UNEP) with the following objectives:

- To assess climate change, the scientific basis of climate change and submit reports thereon so that governments can take steps and frame policies to deal with climate change on scientific basis
- To study the effects of Climate Change
- To study and analyse future risks associated with climate change and
- Develop a suitable approach to combat, mitigate and adapt to climate change

Since its establishment IPCC has completed five assessment cycles and is currently in sixth assessment cycle. IPCC assessments are authored by select leading scientists and the reports are meticulously drafted, reviewed and finalized to provide factsheet and scientific data. It has established three working groups on the following basis

- Physical Science Basis
- Impacts Adaptation and Vulnerability
- Mitigation of Climate Change

In addition, special reports dealing with special issues are also prepared and published so as to create scientific database for taking future steps.

United Nations Framework Convention on Climate Change (UNFCCC)

UNFCCC is an intergovernmental treaty which has been entered into for coping up with the issue of climate change and it provides a framework for dealing with climate change. UNFCCC was opened

for signature in Rio Earth Summit in June 1992 i.e. in United Nations Conference on Environment and Development (UNCED) and it entered into force on 21st March 1994. The main aim of UNFCCC is to stabilize greenhouse gas concentration in a time bound manner so as to reduce and contain the impact of GHG on climate change. UNFCCC recognises and is based on the Doctrine of ‘Common but Differentiated Responsibilities and respective capabilities’. This principle takes into account that the damage to environment and climate as a result of emission of GHG is caused primarily by developed nations since their emission levels and per capita emissions are much higher than developing countries but the adverse impact of the emission of GHG is on the entire community. Therefore, it is the common responsibility of all nations to reduce the emission of GHG but at the same time it recognises special needs of developing and least developed countries to develop and therefore, it is recognized that their level of emission of GHG is likely to increase yet and they are also likely to cause damage to environment and contribute to climate change. The onus is cast on Developed countries which are detailed in Annex A to reduce emission of GHG and also to help developed and developing countries in containing GHG by technology transfer and financial support. UNFCCC mandates preparation of annual inventory of GHG emissions including data for base year (1990). It is worth mentioning here that initially the thrust of UNFCCC was on mitigating the climate change and therefore to contain the emission of GHG. However, after the adoption of Cancun Adaptation Framework (CAF) at Cancun Climate Change Conference, 2010, the focus on adaptation to climate change has also been made. The purpose of CAF is to accord same level of priority to adaptation as is given to mitigation and accordingly to prepare the developing countries to adapt to climate change so as to reduce their vulnerability.

Kyoto Protocol

UNFCCC provided for a framework to deal with climate change. Immediately after adoption of UNFCCC and its entry into force in 1994, it was felt that UNFCCC should be further strengthened by enacting a Protocol which should lay down mandatory targets for reduction of GHG emissions and accordingly, negotiations began to enter into a Protocol. The negotiations resulted in adoption of Kyoto Protocol in December 1997 in Kyoto, Japan. The Protocol entered into force on 16th February 2005. The Protocol has provided mandatory targets for reduction of GHG emissions by Annex 1

countries i.e. developed countries and countries with economies in transition to market economy. The object of the Protocol was to reduce emission of GHG by atleast 5% below 1990 levels during the first commitment period i.e. 2008-2012. The Protocol was based on CBDR and therefore the targets were different for different countries like the target was 8% for European Union, 6% for Canada, 7% for USA and 6% for countries like Hungary, Japan, Poland, New Zealand, Russia, Ukraine etc. Higher targets have been prescribed for second commitment period i.e. for 2013 to 2020 e.g. European Union has agreed to have joint target of 20% reduction of GHG of its 1990 level. The Protocol offers flexibility to the nations to reduce emissions in their own territory or to finance projects in other countries to reduce emissions. Further, the emission targets can be compensated by increasing sinks of carbon dioxide. Oceans and forests are the sinks of carbon dioxide therefore increasing forest cover can reduce increased carbon dioxide from the atmosphere. Protocol has also established Clean Development Mechanism (CDM) which allows flexibility to developed countries to achieve their emission reduction targets by sponsoring emission reduction projects in developing countries and earning saleable Certified Emission Reduction (CER) credits which can be used to achieve reduction targets.

Paris Agreement

Much celebrated Paris Agreement under UNFCCC was entered into in 2015 in 21st Conference of Parties of UNFCCC. The Paris Agreement is viewed as a major breakthrough in climate change efforts wherein the countries of the world came together and agreed for reduction in emissions so as to limit increase in global average temperature upto 2 degree Celsius above pre-industrial levels. The ambitious aim of the Paris Agreement is to contain temperature increase upto 1.5 degree Celsius above pre-industrial levels. Paris Agreement, being entered into under the Framework Convention, also recognized and applied principle of CBDR and therefore, it recognizes that the emissions of developing countries and least developed countries are yet to peak to meet their developmental needs. Therefore, the agreement desires that the global emissions of GHG should peak as early as possible and there should be rapid reductions thereafter. The Agreements calls for ensuring that the global emission of GHG during the period 2050 to 2100 should be to that level only which can be absorbed by natural sinks. Agreement mandated parties to submit Intended National Determined Contributions

(INDC) containing the action plan of the country parties to contain climate change and to reduce the emission of GHG. Agreement provides for revisions of targets periodically i.e. every five years. Like the CAF, it also stresses 'adaptation' strategies so as to adapt to climate change to reduce the vulnerability to the adverse impacts of climate change.

Climate Change Performance Index (CCPI)

After having briefly discussed about climate change, its impact and the international efforts to combat, mitigate and adapt to climate change induced by human activities. It is worthwhile to look at the performance of countries in this direction. Climate Change Performance Index is published annually since 2006 by Germanwatch and Climate Action Network, Europe. Since thirteen years, they are monitoring the efforts of countries responsible for emission of 90% GHGs. The aim of CCPI is to ensure transparency in actions taken by various governments in meeting with their responsibilities regarding climate change and to monitor implementation of international commitments. It is an instrument for measuring efforts of countries regarding GHG emission, energy use, renewable energy, policy measures etc. It was first published in 2006. In November 2017, CCPI, 2018 has been published i.e. thirteenth edition has been published. If we look at the list, we find that first three positions, as usual, have been unoccupied since no country made efforts commensurate with their international commitments. Most of the developed countries such as USA, Russia, Australia, Canada, Japan are in red zone. Only few developed countries like UK and France are in green zone. India has improved its position from 20th spot in CCPI, 2017 to 14th spot in CCPI, 2018.


 Climate Change
 Performance
 Index

Overall Results CCPI 2018

Rank	Country				
1*	–	19.	Brazil	40.	Poland
2.	–	20.	Ukraine	41.	China
3.	–	21.	European Union (28)	42.	Bulgaria
4.	Sweden	22.	Germany	43.	Czech Republic
5.	Lithuania	23.	Belarus	44.	Hungary
6.	Morocco	24.	Slovak Republic	45.	Algeria
7.	Norway	25.	Luxembourg	46.	Argentina
8.	United Kingdom	26.	Romania	47.	Turkey
9.	Finland	27.	Mexico	48.	South Africa
10.	Latvia	28.	Egypt	49.	Ireland
11.	Malta	29.	Cyprus	50.	Japan
12.	Switzerland	30.	Estonia	51.	Canada
13.	Croatia	31.	Slovenia	52.	Malaysia
14.	India	32.	Belgium	53.	Russian Federation
15.	France	33.	New Zealand	54.	Chinese Taipei
16.	Italy	34.	Netherlands	55.	Kazakhstan
17.	Denmark	35.	Austria	56.	United States
18.	Portugal	36.	Thailand	57.	Australia
		37.	Indonesia	58.	Republic of Korea
		38.	Spain	59.	Islamic Republic of Iran
		39.	Greece	60.	Saudi Arabia

 Source : <https://germanwatch.org>

Climate Change and Indian Efforts

As discussed earlier, the Kyoto Protocol has set binding targets for Annex I countries. India is a non-annex I country and hence had no binding targets in the first commitment period. Further, it can not be lost sight of that India is a developing countries and millions of people still does not have access to electricity. Thus the developmental needs of the country coupled with energy needs are likely to increase India's GHG emissions in the decades to come. However, India is also prone to adverse effects of climate change including subsidence of coastal areas. Therefore, India has been participating

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in climate change programmes and fulfilling its commitments under the international law. India has actively participated in Clean Development Mechanism (CDM) established under Kyoto Protocol. India has also ratified Second Commitment Period targets under the Kyoto Protocol in August 2017. A brief analysis of India's efforts in this direction is given hereinafter.

India has not enacted a legislation to deal with climate change however, Government of India constituted Prime Minister's Council on Climate Change in June 2007. The main aim of this Council is to advise the government regarding climate change policies and effects. The objectives of PM's Council on Climate Change are as under:

- To coordinate National Plan for assessment, adaptation and mitigation of Climate Change
- To advise the government
- To facilitate inter-ministerial coordination

Since its establishment, the Council has been reconstituted by successive governments to advise the national government regarding climate change action plan and for adaptation and mitigation of climate change. The Council is chaired by the Prime Minister and various Union ministers like Finance Minister, External Affairs Minister, Power Minister, Minister of Environment, Forests and Climate Change, Minister of Agriculture etc are members alongwith members from TERI (The Energy and Resource Institute) and BEE (Bureau of Energy Efficiency).

National Action Plan on Climate Change

India has also launched a National Action Plan on Climate Change (NAPCC) in 2008. The Action Plan consists of eight national Missions which are briefly discussed hereunder:

- Jawahar Lal Nehru National Solar Mission – Mission was launched in 2008 to establish India as a global leader in solar energy to contribute towards clean energy and clean development. The aim of the mission is to develop 20000 MW of solar energy. To facilitate the generation of solar energy, 20 million square meter thermal collector area is required. The country has also established Clean Energy Fund in 2010 and levied cess on Coal i.e. Clean Energy Cess which

has been renamed as Clean Environment Cess in 2017. The purpose of this fund is to finance clean energy project and promote clean energy. To further contribute towards clean energy, Electricity Act, 2003 has mandated purchase of grid based power from renewable sources.

- National Mission for Enhanced Energy Efficiency – The aim of this mission is to ensure energy efficiency improvement in large energy intensive industries and to facilitate Energy Efficient Economic Development. For this purpose, country is focussing on development and use of energy efficient appliances and equipments e.g. super efficient ceiling fans have been developed. India has also been promoting and distributing LED bulbs so as to promote energy efficiency and thereby lower energy consumption. Target is to save fuel and avoid capacity addition.
- National Mission on Sustainable Habitat – The target of this mission is to make cities sustainable by improvements in energy efficiency and by promoting use of public transport. The chief features of this mission are
 - Extension of Energy Conservation Building Code, 2007 to all new and existing buildings
 - To optimize energy demands of large commercial buildings
 - Shift to energy efficient and convenient Public Transport
 - Utilization of Urban Waste to derive fuel – Refuse derived Fuel (RDF)
- National Water Mission – One of the components of NAPCC is national water mission. The aim of this mission is
 - To Conserve Water, minimize wastage and equitable distribution of water
 - Creation of Ground Water Monitoring Wells
 - Development of water database
 - India has made efforts for revision of National Water Policy, 2012 in line with these aims
- National Mission for Sustainable Agriculture – This mission aims at developing sustainable agriculture as a climate resilient system and at the same time to ensure food security which is an important need of developing country like India. For the purpose, the mission promotes

- On farm water use efficiency
 - Soil health management
 - Development of degraded land and
 - Micro irrigation
- National Mission for Sustaining the Himalayan Ecosystem – Himalayan Ecosystem is very important not only for India but for the world at large and it is important that adequate efforts are made to sustain the Himalayan Ecosystem. For this purpose, this mission has promoted following steps
 - Sustaining and safeguarding Himalayan glaciers and mountain ecosystems
 - Protection of Biodiversity, wildlife of Himalaya
 - Identification of institutions for studies on Himalayan ecosystems
 - Creation of centres for bridging knowledge gaps
 - Identification and training of experts
 - Creating and strengthening Observational network
 - National Mission for a Green India – Green India is an important mission on which both the Union and State governments have been working in cohesion. States have been asked to make State level plans for the same and many states have submitted and are implementing their plans. The aim of this mission in consonance with National Forest Policy is to increase the forest cover on 5 million hectares of forest and non-forest land and also to increase forest based livelihood.
 - National Mission on Strategic Knowledge for Climate Change – The purpose of this mission is to promote research and development on climate change and the likely impact of climate change on India so as to make mitigation and adaptation strategy accordingly. The mission aims at providing and ensuring funding for research on climate change.

The PM's Council on Climate Change and Ministry of Environment, Forests and Climate Change is working on strengthening and revamping NAPCC and it is expected to launch three more missions under NAPCC on Impact of Climate Change on Health, Coastal Zones and regarding Waste to Energy (The Hindu, 2017).

Conclusion

From the foregoing discussion, it is clear that mankind has forced climate change due to their unregulated and rampant industrial and other activities. After industrialisation, there has been increase in global surface air temperature and it is likely to continue in future also. The main reason for anthropogenic climate change is the emission of GHGs by humans whose concentration is forcing the climate change including global warming which is going to affect a vast majority of world population including India. If the pace of climate change continues like this, it would adversely affect the survival of humans in various parts of the world especially coastal areas. Accordingly, recognising these adverse effects, international community entered into UNFCCC and thereunder Kyoto Protocol was entered into in 1997 which became effective in 2005. Binding targets for reduction of GHGs by developed countries were fixed in the Kyoto Protocol. Paris Agreement has also been entered into in 2015 with the ambitious target of containing temperature increase upto 1.5 degree Celsius of pre-industrial levels. However, despite the international instruments, the performance of the developed countries is far from satisfactory. India has also made its efforts regarding mitigating and adapting to climate change though Indian efforts also fall short of the requisite level e.g. Clean Environment Cess though has been levied and huge funds are collected thereunder but as per CAG Reports, the fund has not been utilized for promoting and financing Clean energy.

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